

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 31 January 2025

Team ID: LTVIP2025TMID35065

Project Name: Trafficelligence: Intelligent Traffic Volume Prediction System

Maximum Marks: 4 Marks

1. Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	1. Registration through form (email + password + confirm) 2. Registration through Gmail OAuth 3. Registration through LinkedIn OAuth
FR-2	User Confirmation	1. Confirmation via email link 2. Confirmation via SMS OTP
FR-3	Login & Authentication	1. Login using email/password 2. Social login via Gmail/LinkedIn 3. Password reset via email link
FR-4	Traffic Prediction Dashboard	1. View real-time prediction graphs 2. Filter by region, date, and time 3. Export data as CSV/PDF 4. Configure threshold alerts for high traffic
FR-5	Admin & Dataset Management	1. Admin user management (add/edit/delete users) 2. Upload new traffic datasets (CSV/API) 3. Monitor model performance via admin dashboard

2. Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, intuitive UI with guided workflows; help tooltips and mobile-responsive design.
NFR-2	Security	Secure authentication (hashed passwords, OAuth 2.0), HTTPS encryption, role-based access control.
NFR-3	Reliability	≥ 99% system uptime; automatic retries for failed data processes; fallbacks for model errors.
NFR-4	Performance	Predictions served within 2 seconds; dashboard updates within 3 seconds; optimized queries and caching.
NFR-5	Availability	Geo-redundant cloud hosting; regular backups for user and prediction data; 24×7 monitoring.

NFR No.	Non-Functional Requirement	Description
NFR-6	Scalability	Horizontal scaling for traffic volume surge; ability to plug in new data sources via modular APIs.